## Summary of this week

- At the end of every week, we quickly summarize progress made during the week and preview upcoming topics.
- This week, we introduced the purpose and basic features of Kalman filters.
  - You learned that the purpose of a KF is to estimate the state of a dynamic system.
  - You learned that the KF requires a model of system dynamics and noises.
  - You learned that the KF repeatedly performs two steps: prediction and correction.
  - You learned that the output of a KF is an estimate of the model's state as well as the uncertainty of the estimate.
  - You learned some example application categories that use KF.





## Where to from here?

- You are now ready to study the background topics that are necessary to be able to understand more deeply how a KF works, and to implement it for your application.
- Next week, we focus on learning about state-space models.
  - Why do we need to know about them?
  - What are some example model types that we might want to use for some applications?
  - How do we think about the time-domain response of a state-space model?
  - How do we convert a continuous-time model to a discrete-time model, and how do we simulate them?
  - Can the model tell us if it is even possible for a KF to estimate its state?





## Credits

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